

TECHNICAL AND FINANCIAL OFFER

Steel-framed sandwich-panel roof over the existing roof slab
Administration Building, Regatta Tourist Village, Tripoli

Offer No.	RTV-2026-01 Rev D (supersedes earlier issues)	Date	29 September 2026
To	Regatta Tourist Village - Administration	Valid until	14 October 2026
From	Eng. MALEK ABOZRAIG, malek.abozraig@gmail.com	Design basis	Design Rev 8a (reduced roof area, architect's plan A_103)

Dear Sirs,

Further to the survey of the existing administration building, the structural design completed for it and the architect's 2nd-level layout of 29 September 2026, we are pleased to submit our revised offer for the supply, fabrication and erection of a steel roof structure with insulated sandwich-panel cladding over the existing roof slab, as described below and shown on the attached framing plan and 3D view. The offer is a lump sum covering all materials, workshop fabrication, transport, site works and erection listed in the scope, delivered as a complete and weather-tight roof.

1. The structure

The new roof is a light steel post-and-beam structure standing on the existing 300 mm roof slab, with one steel column on each of the 23 existing concrete columns under the enclosed area, so that all loads pass straight into the building's own structure. Level primary beams run along the column rows; sloping rafters, continuous over the middle row, sit on top of them and carry cold-formed purlins and 50 mm insulated PIR sandwich panels on a single roof plane falling 6 % to the north, where two box gutters and four downpipes on the north facade take the rainwater away. Lateral stability is provided by six X-braced wall bays and a braced roof plane, designed for the Tripoli wind and seismic conditions. The stair well remains open with upstands and flashings; the south-west terrace and the skylight terrace are not roofed, and the roof edges sit on the existing column lines K19-K20 and K16-K17-K18. The roofed area is 316 m2; the clear height under the steel is 3.0 m at the low north eave and about 4.0 m at the south edges.

Element	Specification
Columns	23 no. HEA 140, S275, one over each existing concrete column under the roof
Primary beams	IPE 200, S275, level, bolted cap plates on the columns
Rafters and trimmers	IPE 180 continuous over the middle primaries, S275, single roof plane at 6 % falling north, bolted seats on the primaries
Purlins	Z 200 x 2.0 cold-formed, S350GD galvanised, at 1.5 m centres (wall girts excluded)
Bracing	L 60 x 6 wall diagonals in 6 bays; M20 roof rods in 12 panels
Bases	4 post-installed dia 16 rebars per column into the existing column head, grouted shear keys, 20 mm plates, 25 mm grout bed
Roof cladding	PIR sandwich panels 50 mm, colour-coated steel both faces, with ridge, eave and verge flashings and well upstands
Rainwater	Two box gutters 150 x 100 with overflows, four 100 mm downpipes on the north facade
Steel weight	about 11 tonnes, delivered blast-cleaned; corrosion protection is not included (see sections 3 and 7)

2. Scope of supply and works

- Detailed fabrication drawings and the bill of materials based on the approved design (Rev 8a, reduced roof area).
- Supply of all structural steel sections, plates, bolts, rebars, anchors and cold-formed purlins and girts.
- Workshop fabrication, blast cleaning of the steel, transport to site and offloading.
- Site survey of the slab top, scanning of the 23 column heads, drilling and injection of the anchors, coring and grouting of the shear keys, and proof tests on three bars.
- Erection of columns, beams, rafters, bracing and purlins with all connections, including the temporary bracing needed during erection.
- Supply and fixing of the roof sandwich panels with all flashings, well upstands, sealants and fasteners.
- Supply and fixing of gutters, downpipes and the associated brackets and flashings.
- Site management, safety measures, cleaning of the site and handover with as-built drawings.

3. Not included

- Corrosion protection of the steelwork (paint system or galvanising) is not within the scope of this offer; the recommended Jotun system is given in section 7 and can be offered separately.
- Wall cladding of any kind and its girts or fixings (the architect's drawings show 20 cm gypsum walls; to be offered separately once the wall system is selected).
- Verification of the existing building (as-built drawings, slab-top survey drawing, adequacy statement of the existing columns and foundations) and any strengthening it may require.
- Building permits, authority fees, independent checking and any works to the existing slab or facade beyond the anchors, keys and downpipe fixings.
- Electrical, mechanical, lighting, ceilings, floors and finishes under the new roof.

4. Programme

The works are planned over about three weeks from receipt of the advance payment, subject to site access and to the scan results at the column heads.

Week	Activity
1	Fabrication drawings approved; slab survey and column-head scans; steel and panel procurement; workshop fabrication starts
2	Fabrication completed and delivered; anchors drilled and injected, shear keys cored and grouted; columns and beams erected
3	Rafters, bracing and purlins completed; roof panels, flashings, gutters and downpipes fixed; cleaning and handover

5. Price

Lump-sum price for the complete scope in section 2	LYD 213,000
Libyan Dinars, fixed for the validity period, inclusive of materials, fabrication, transport, site works, erection and supervision. Taxes and fees, if any, are for the client's account.	

6. Payment terms

Stage	Share	Amount, LYD
Advance on signing the order; starts the programme	50 %	106,500
On start of erection on site	25 %	53,250
On completion and handover	25 %	53,250

7. Conditions

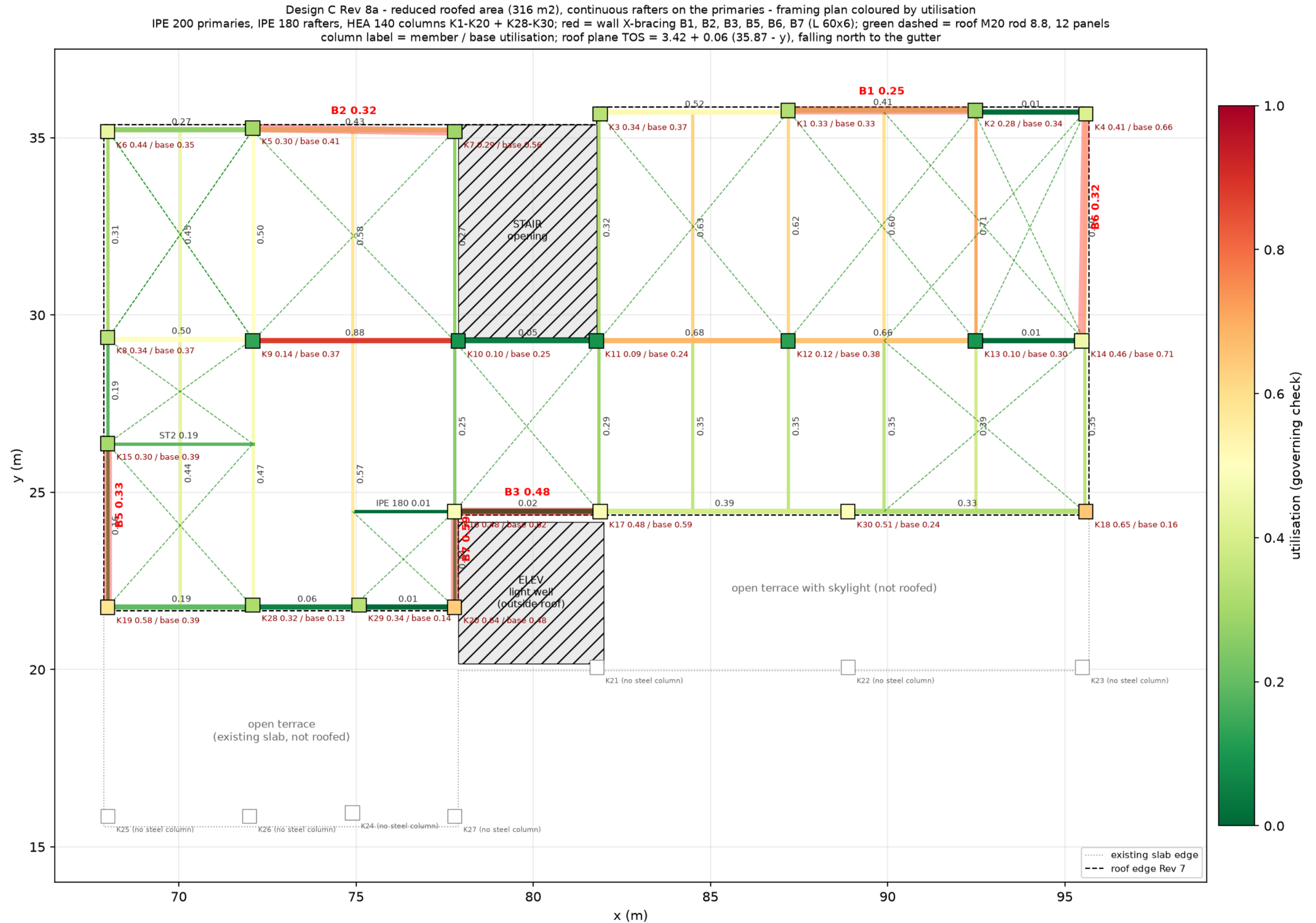
- Validity: this offer is valid for 15 days from its date, until 14 October 2026, because of the movement of steel prices.
- Warranty: 12 months from handover on the steel structure, panel fixings and workmanship; panel warranties as per the manufacturer.
- Corrosion protection (recommendation, not included): the site is 0.3-2 km from the sea, corrosivity category C5 (ISO 12944-2). We recommend a Jotun three-coat system applied in the workshop over Sa 2 1/2 blast cleaning: Barrier 80 zinc-rich epoxy primer 60 microns, Jotamastic 90 epoxy mastic 150 microns, Hardtop XP polyurethane topcoat 60 microns, total 270 microns dry film, ISO 12944-5 high durability (over 15 years); site touch-up of bolts, welds and damage with Jotamastic 90 and Hardtop XP. Hot-dip galvanising to EN ISO 1461 plus Hardtop XP is the alternative. This can be priced on request.
- The design is based on the assumptions listed in the design report (Rev 8a) and the load basis Rev 3. The client provides as-built information on the existing building and free access to the roof and to the north facade for the downpipes; anything found at the column heads that differs from the assumptions (bar positions, concrete quality, slab thickness) is dealt with by the fallback base details of the design set and, if it changes the scope, by a written variation before work continues.
- The client confirms that the six braced wall bays shown on the framing plan (K1-K2, K5-K7, K16-K17, K15-K19, K4-K14, K16-K20) remain free of doors and windows, and that the architect adjusts the 2nd-level wall lines to the roof edges shown.
- Materials: structural steel S275 to EN 10025, bolts 8.8, cold-formed purlins S350GD, PIR panels with a manufacturer's certificate; anchor system with a European Technical Assessment for post-installed rebar.
- Design standards: Eurocodes EN 1990, 1991, 1993 and 1998 with parameters for Tripoli, as recorded in the design report.
- Any change to the roof plane, the openings, the cladding type or the scope after the order is priced separately.
- The offer excludes anything not expressly stated in section 2.

8. Attachments

- Attachment A: roof framing plan, design Rev 8a (A3).
- Attachment B: 3D view of the steel structure, design Rev 8a.
- The detailed drawing set (fabrication and erection drawings, A2) is produced after the order on the approved Rev 8a design; the Rev 6a set RTV-ST-S00 to S06 remains available as reference for the typical details.

We remain at your disposal for any clarification and look forward to working with you on this project.

For the offer	For the client (acceptance)
Eng. MALEK ABOZRAIG malek.abozraig@gmail.com Date:	Name: Position: Date:



Steel roof, design Rev 8a - reduced roofed area (316 m2): view from the south-west, roof panels hidden

